THE UNIVERSITY OF DODOMA

COLLEGE OF INFORMATICS AND VIRTUAL EDUCATION

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CSE)

FINAL YEAR PROJECT PROPOSAL

2025/2026 ACADEMIC YEAR

TITLE:

Smart Poultry Health Monitoring and Automated Management System Using IoT and AI

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2. Problem Statement

Poultry farming faces significant challenges in maintaining optimal health conditions and preventing disease outbreaks, leading to substantial economic losses and reduced productivity. Traditional poultry management methods suffer from several critical issues:

- **High Mortality Rates:** Disease outbreaks often go undetected until significant bird loss occurs, with respiratory infections causing up to 30% mortality in untreated flocks
- **Labor Intensive:** Manual monitoring of environmental conditions, feeding, and watering requires constant human intervention, increasing operational costs
- **Delayed Disease Detection:** Farmers typically identify health issues through visible symptoms, by which time diseases have already spread through the flock
- **Inconsistent Environmental Control:** Manual temperature and humidity regulation often leads to fluctuations that stress birds and compromise immune systems
- **Poor Resource Management:** Inefficient feeding and watering practices result in wastage and increased production costs
- **Limited Real-time Monitoring:** Lack of continuous health and environmental tracking prevents proactive intervention

Small-scale poultry farmers in particular lack access to affordable, automated solutions for comprehensive flock management, relying on reactive rather than preventive approaches to health management.

3. Main Objective

To design and develop an integrated IoT-based poultry monitoring system that automatically tracks environmental conditions, detects early signs of disease through behavioral patterns, and manages feeding and watering processes to optimize poultry health and productivity while reducing human intervention.

4. Specific Objectives

i. To design and implement a comprehensive sensor network for real-time monitoring of critical environmental parameters including temperature, humidity, ammonia levels, and bird activity within the poultry coop.

ii. To develop an automated resource management system that intelligently controls feeding, watering, and environmental regulation based on sensor data and predefined thresholds.

iii. To create an AI-powered health monitoring algorithm that analyzes behavioral patterns and environmental data to detect early signs of common poultry diseases including respiratory infections, heat stress, and general sickness.

iv. To build an alert and notification system that provides immediate warnings to farmers through multiple channels (LCD display, mobile notifications) when abnormal conditions or potential health risks are detected.

v. To develop a user-friendly monitoring dashboard that displays real-time system status, historical trends, and health analytics for informed decision-making.

vi. To validate system performance through controlled testing that measures accuracy in environmental regulation, disease detection reliability, and resource utilization efficiency.

5. Significance

Economic Impact

- **Reduced Operational Costs:** Automation decreases labor requirements by approximately 60%, significantly lowering production expenses
- **Minimized Mortality Losses:** Early disease detection can reduce bird mortality rates by 25–40%, preserving farm revenue
- **Optimized Resource Utilization:** Smart feeding and watering systems can decrease feed wastage by 15–20% and water consumption by 10–15%
- **Improved Productivity:** Healthier birds demonstrate better growth rates and higher egg production, increasing overall farm output

Agricultural Innovation

- **Democratizing Precision Agriculture:** Makes advanced monitoring technology accessible and affordable for small-scale poultry farmers
- **Knowledge Advancement:** Contributes to understanding the relationship between environmental factors and poultry health through data-driven insights
- **Sustainable Practices:** Promotes efficient resource use and reduces environmental impact through optimized feeding and waste management

Social Benefits

- **Food Security Enhancement:** By reducing poultry losses, the system contributes to more stable protein supply chains
- **Rural Employment Support:** Helps sustain small-scale poultry operations that provide livelihoods in rural communities
- **Farmer Education:** Provides valuable insights into poultry health management, improving overall farming practices

Technical Significance

- **IoT Integration Exemplar:** Demonstrates practical implementation of Internet of Things in agricultural applications
- **Edge Computing Application:** Showcases real-time data processing at the device level for immediate response
- **AI in Agriculture:** Illustrates how machine learning can transform traditional farming practices through predictive analytics

Educational Value

- **Cross-disciplinary Learning:** Integrates concepts from electronics, programming, data science, and agricultural science
- **Research Foundation:** Establishes baseline for future innovations in smart livestock management systems
- **Technology Transfer:** Bridges the gap between academic research and practical agricultural applications

This project addresses critical challenges in poultry farming while demonstrating the transformative potential of IoT and AI technologies in transforming traditional agricultural practices into smart, efficient, and sustainable operations.